Execute code at some point in the future (medium)

We want to respond to an operation that happens in our system, but we want to do so minutes or hours in the future. We need to use an async job to schedule code to execute at that point in the future. This is called a scheduled job.

Optimize repeating task (advanced)

An operation happens frequently in our system, and it results in an expensive operation that calculates a cache value. We could let it calculate as many times as the operation happens, but that would be wasteful because it occurs frequently. Instead, we can leverage a unique job that ensures the operation only occurs once every so often.

Robust data pipeline (advanced)

We have a data pipeline that requires a batch of jobs to finish before the next step occurs. That step produces many more tasks, which have to complete before the final task occurs. This is called a workflow and is fairly complex to implement. Job systems can help us by allowing us to tap into existing abstractions to write this complex data pipeline.

These are only a few use cases for async jobs. There are so many possibilities, and you'll start seeing more of them once you get comfortable with an async job system.

Let's go over a litmus test that you can use to determine whether a piece of code is a good candidate for an async job.

Async Job Litmus Test

Consider the following questions when evaluating a piece of code to see if it can be made async:

Is this code inherently costly?

Sending HTTP requests, interacting with SMTP servers, and processing data-intensive requests are all things that have an inherent cost to them. If you execute this type of code in the main controller, it can hold up the connection and prevent a response from being received by the user. Instead, consider extracting it into an async job and allow it to finish behind the scenes.

Is this code at a relatively-high risk of failure?

Certain pieces of code are more likely to experience intermittent failures that have nothing to do with the operation itself. For example, sending an HTTP request to a service that's experiencing an outage